Rental Vehicle Manager

A Java Swing application for managing a small vehicle rental system. This project demonstrates object-oriented programming, inheritance, enums, array-based data management, GUI development, and event handling. It is designed as a final course project and is compatible with Eclipse IDE.

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Overview

Vehicle Manager allows users to add, remove, search, sort, rent, return, and view vehicles through a graphical user interface. The program supports general vehicles, cars, and trucks. Each vehicle stores information such as license plate, manufacturer, year, color, price, number of seats, status, gas tank size, and reservation data.

The application uses a manager/controller class to store and control the vehicle array, while the GUI acts as the front end for user interaction. Cars and trucks extend the parent Vehicle class, which helps avoid repeated code and shows inheritance in practice.

Main Java concepts illustrated:

- Classes and encapsulation using Vehicle, Car, Truck, and VehicleManager
- Inheritance through Car extends Vehicle and Truck extends Vehicle
- Enums for Manufacturer, Color, Status, and Type
- Arrays for storing vehicles and reservations
- Swing GUI components such as JFrame, JPanel, JTextField, JComboBox, JButton, JLabel, and JTextArea
- Event handling with button action listeners
- Basic input validation and error messages using JOptionPane

Key Features

1. **Add Vehicles –** Input vehicle information and create either a car or a truck.
2. **Remove Vehicles –** Remove a vehicle from the system using its license plate.
3. **Search Vehicles –** Search for vehicles by license plate or other available search options in the GUI.

4. **View Vehicles –** Display all stored vehicles in the GUI display area.
5. **Rent Vehicles –** Add a customer name to the vehicle reservation list.
6. **Return Vehicles –** Remove the current reservation and move the next customer up if one exists.
7. **Sort Vehicles –** Sort vehicles by availability and price so available/cheaper options are easier to find.
8. **Update Vehicle Data –** Update status, price, or mileage-related information when supported by the manager and GUI.
9. **Validation –** Handle invalid number inputs and missing fields with dialog messages.

Requirements

- Eclipse IDE for Java Developers
- JDK 17 or higher

Run the Program

Step 1 — Import or Create the Project

10. Open Eclipse IDE.
11. Go to File → New → Java Project.
12. Enter a project name, such as VehicleManager.
13. Click Finish.
14. Copy the .java files into the project folder.
15. Make sure all files use the same package declaration: package rentalcars;

Step 2 — Required Java Files

- Vehicle.java
- Car.java
- Truck.java
- VehicleManager.java
- VehicleManagerFE_GUI.java
- Manufacturer.java
- Color.java
- Status.java
- Type.java

Step 3 — Run the Program

16. Locate the main GUI file: VehicleManagerFE_GUI.java.
17. Right-click the file.
18. Choose Run As → Java Application.
19. The Vehicle Manager GUI window should open.

Step 4 — Optional: Export as Runnable JAR

20. Right-click the project → Export... → Runnable JAR File.
21. Choose the launch configuration for VehicleManagerFE_GUI.
22. Select *Package required libraries into generated JAR*.
23. Click Finish.

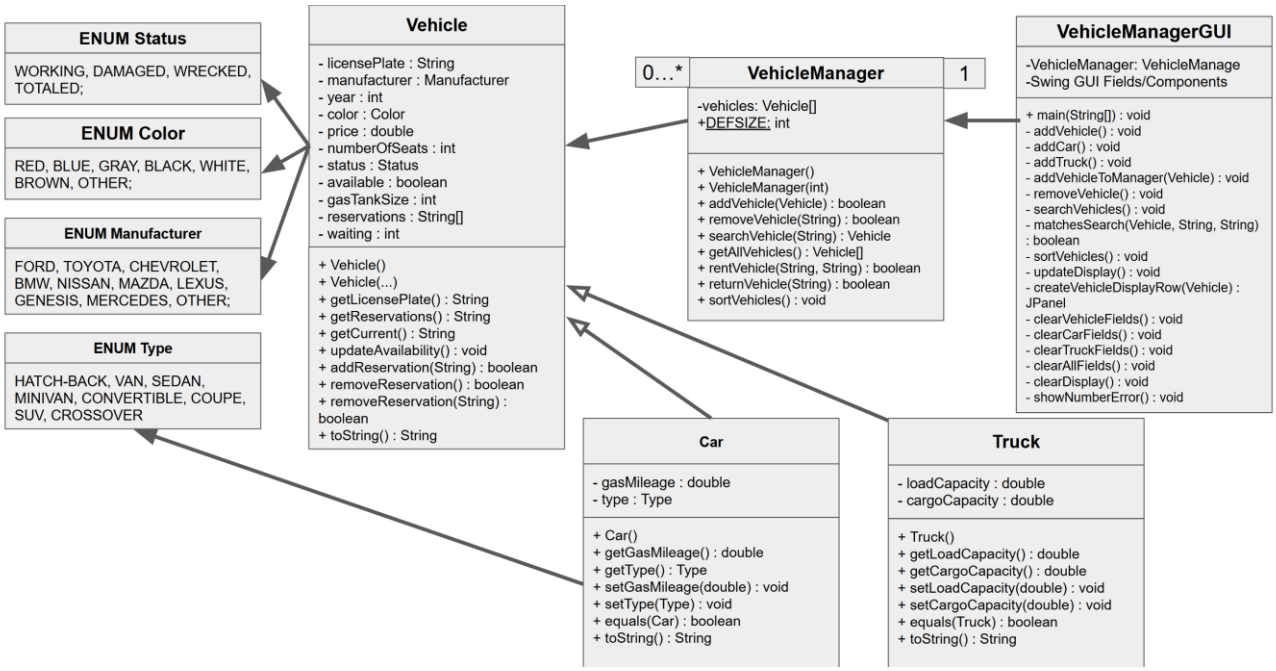
How to Use

- Launch the app from Eclipse.
- Enter vehicle information in the Vehicle Information panel.
- Choose whether the vehicle is a Car or Truck.
- For cars, enter gas mileage and select a vehicle type.
- For trucks, enter load capacity and cargo capacity.
- Click Add Vehicle to save the vehicle into the manager.
- Use the search/remove field to search for or remove a vehicle by license plate.
- Use renter name and license plate fields to rent or return a vehicle.
- Use Show All or the display area to view stored vehicle records.
- Use invalid values such as letters in number fields to test input validation.

Project Structure

File	Purpose
Vehicle.java	Parent class that stores common vehicle information and reservation logic.
Car.java	Child class that adds car-specific fields such as gas mileage and vehicle type.
Truck.java	Child class that adds truck-specific fields such as load capacity and cargo capacity.
VehicleManager.java	Backend manager/controller that stores vehicles in an array and performs add, remove, search, rent, return, sort, and update operations.
VehicleManagerFE_GUI.java	Front-end Swing GUI that allows the user to interact with the vehicle manager.
Manufacturer.java	Enum for vehicle manufacturer values.
Color.java	Enum for vehicle color values.
Status.java	Enum for vehicle condition/status values.
Type.java	Enum for car/body type values.

UML Diagram



Testing Checklist

- App compiles and runs successfully.
- GUI window opens without errors.
- Add Vehicle works for both Car and Truck.
- Remove Vehicle works using the correct license plate.
- Search displays matching vehicles correctly.
- Show All displays every vehicle currently stored.
- Rent Vehicle adds a reservation and updates availability.
- Return Vehicle removes the current reservation and updates availability.
- Sort places vehicles in the expected order.
- Invalid number input is handled gracefully.
- Blank required fields do not crash the program.
- GUI display updates after each action.

Future Enhancements

- Save and load vehicle data using CSV, JSON, or text files.
- Improve search and filtering with more dropdown options.
- Add an edit screen for changing all vehicle details.
- Add images/icons for car and truck entries.
- Add stronger validation for license plate format, year range, price, and gas tank size.
- Add theme customization or a more modern GUI layout.

Credits

Developed by Group 4

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Course: CSE 1325 – Object-Oriented Programming

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